



PHYSICS HSSC-I

SECTION – A (Marks 17)

Time allowed: 25 Minutes

Section – A is compulsory. All parts of this section are to be answered on this page and handed over to the Centre Superintendent. Deleting/overwriting is not allowed.

Do not use lead pencil.

حصہ اول لازمی ہے۔ اس کے جوابات اسی صفحہ پر دے کر تمام مرکز کے حوالے کریں۔ کاٹ کر دوبارہ لکھنے کی اجازت نہیں ہے۔ لید پینسل کا استعمال نہ کریں۔

Version No.				
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ROLL NUMBER						

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Answer Sheet No. _____

Invigilator Sign. _____

Fill the relevant bubble against each question according to curriculum:

Candidate Sign. _____

Question	A	B	C	D	A	B	C	D
1. The unit of force in terms of base units:	kg.m/s	kg.m/s ²	kg.m ² /s	kg ² .m/s ²	☐	☐	☐	☐
2. The dimension $[ML^{-1}T^{-2}]$ represents:	Pressure	Force	Power	Work	☐	☐	☐	☐
3. Torque $\vec{\tau}$ on a body is given by:	$\vec{\tau} = \vec{m} \times \vec{a}$	$\vec{\tau} = \vec{F} \times \vec{d}$	$\vec{\tau} = \vec{r} \times \vec{F}$	$\vec{\tau} = \vec{l} \times \vec{\omega}$	☐	☐	☐	☐
4. For a freely falling body, the vertical velocity at the 4 th second is:	94.9 ms ⁻¹	47.5 ms ⁻¹	40 ms ⁻¹	39.2 ms ⁻¹	☐	☐	☐	☐
5. The maximum range of a projectile is obtained when launch angle $\theta =$	30°	45°	60°	90°	☐	☐	☐	☐
6. The SI unit of angular velocity is:	Rad	N.m	m/s ²	rad/s	☐	☐	☐	☐
7. The centripetal force for a body of mass m moving with speed v in a circle of radius r is:	mv	mv ² r	$\frac{mv^2}{r}$	$\frac{m}{r^2}$	☐	☐	☐	☐
8. The work done by a gas at constant pressure is given by:	P. ΔV	$\frac{\Delta V}{P}$	ΔV	V. ΔP	☐	☐	☐	☐
9. The first law of thermodynamics is expressed as:	Q = U + V	Q = T + S	Q = ΔU + W	Q = mL	☐	☐	☐	☐
10. The Doppler effect explains the change in:	Frequency	Amplitude	Speed of light	Wavelength only	☐	☐	☐	☐
11. For constructive interference, path difference Δx should be:	n λ	$\frac{(2n+1)\lambda}{2}$	$\frac{(n+1)\lambda}{2}$	$\frac{n\lambda}{2}$	☐	☐	☐	☐
12. Electric field intensity E is defined as:	F/d	F/q	V/l	W/q	☐	☐	☐	☐
13. A 10 Ω resistor is connected across the terminals of a 10V battery. The power dissipation of the resistor is:	100 W	50 W	10 W	25 W	☐	☐	☐	☐
14. Magnetic flux Φ through a surface area A is:	B/A	B.A	A/B	B/A ²	☐	☐	☐	☐
15. Lenz's law states that induced emf opposes:	Current	Resistance	Power factor	Change causing it	☐	☐	☐	☐
16. An inertial frame of reference is the one:	Which is at rest	Which has zero acceleration	Which is under a force	Which has variable acceleration	☐	☐	☐	☐
17. At what speed the mass of a body becomes double of its proper mass?	$\frac{2}{3}c$	$\frac{\sqrt{3}}{2}c$	$\frac{\sqrt{2}}{3}c$	$\frac{1}{3}c$	☐	☐	☐	☐

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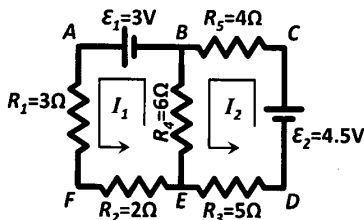
• $P = \frac{V^2}{R} = I^2 R = V \times I$ • $g = 9.8 \text{ ms}^{-2}$ • $F = ma$ • $m = \frac{m_0}{\sqrt{1 - \frac{v^2}{c^2}}}$ • $v_f = v_i + gt$

**PHYSICS HSSC-I****Time allowed: 2:35 Hours****Total Marks Sections B and C: 68****SECTION – B (Marks 42)****Q. 2 Answers the following parts briefly.****(14 x 3 = 42)**

(i)	Derive the relation for centripetal acceleration (a_c) of a body moving in a circle of radius r with velocity v .	03	OR	Find the power delivered by the engine for attaining velocity (2,3) m/s, while it exerts a force (7,-3) N.	2+1
(ii)	What is meant by dot product of two vectors? Give mathematical expression and one example.	1x3	OR	Derive $PV=nRT$ for an ideal gas.	03
(iii)	Differentiate between constructive and destructive interference of waves.	1.5+ 1.5	OR	A rabbit jumps over an object 0.7 m high: a. Considering just its vertical motion, calculate its vertical speed when it leaves the ground. b. How long a time it stays in the air?	1+2
(iv)	Briefly describe the terms Stress, Strain, and Young's modulus.	03	OR	Briefly explain the applications of Doppler Effect in radar and astronomy.	1.5+ 1.5
(v)	What is meant by elastic potential energy? Give its mathematical form.	2+1	OR	A projectile is launched horizontally from the top of a 50 m high cliff and lands at a distance of 20 m from the base of the cliff. Determine the magnitude of the launch velocity.	03
(vi)	The energy of photon is given by $E=hf$, find the dimensions of Planck's constant h , where f is frequency.	03	OR	A point source of sound emits sound energy at a rate of 15 W. Calculate the intensity of sound at a distance of 4 m from the source.	03
(vii)	A ball of mass 120 g is released from a height of 30 m. if the ball encounters an air resistance of 0.3 N, find the kinetic energy of the ball just before striking the ground.	03	OR	Briefly describe the phenomenon of polarization in transverse waves.	03
(viii)	Briefly describe the elastic deformation and plastic deformation.	03	OR	Calculate the electric field intensity at a distance of 0.5 m from a point charge of 2×10^{-6} C in a vacuum.	03
(ix)	A current of 2 A is flowing in a copper conductor with a cross-section of 1 mm^2 . Find the drift velocity of the electrons. (For copper, $n=8.5 \times 10^{28}$ per m^3)	03	OR	Derive the formula for Kinetic Energy [using the equations of motion].	03
(x)	Find the efficiency of a heat engine operating between 427°C and 27°C .	03	OR	The electric field at a point due to a point charge is 26 N/C and the electric potential at that point is 13 J/C . Calculate the distance of the point from the charge.	03
(xi)	Determine the relationship for maximum height of a projectile.	03	OR	Differentiate between conservative and non-conservative forces.	03
(xii)	An object having a net charge of $24 \mu\text{C}$ is placed in a uniform electric field of 610 N/C directed vertically. What is the mass of this object if it floats in the field?	03	OR	A proton is moving in a circular orbit of radius 12 mm in a uniform 0.30 T magnetic field perpendicular to the velocity of the proton. Find the speed of the proton.	03
(xiii)	Find the momentum of a positron, moving with relativistic speed. The energy of the positron is 0.47 MeV .	03	OR	According to which principle ships float on water? Explain briefly.	03
(xiv)	Distinguish between crystalline and amorphous solids.	1.5+ 1.5	OR	Calculate the resistance of wire 15 m long that has a diameter of 1.5 mm and resistivity of $2.63 \times 10^{-2} \Omega \text{ m}$.	1+2

SECTION – C (Marks 26)**Attempt the following questions.****(Use of graph paper is not necessary. Candidates can make their own grid on answer book if required)**

Q.3	What is meant by angular momentum? Determine the angular momentum of a rigid body.	2+5	OR	State and explain the second law of thermodynamics. Also draw diagrams.	3+2 +2
Q.4	Explain the origin of the forces between current-carrying conductors and determine the direction of the forces.	4+2	OR	How does the frequency of sound of a source appear when it moves: a. Towards a stationary observer b. Away from a stationary observer	4+2
Q.5	Explain the working of venturi meter. Give its mathematical form. Draw its sketch.	07	OR	What is meant by quark? Explain the quark family of particle.	2+5
Q.6	Find Current flowing through each resistor, using Kirchhoff's law, in the given circuit.	06	OR	What is special theory of relativity? Describe the following consequences of special theory of relativity: a. Mass variation b. Length contraction c. Time dilation	2+2 +2



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$$\begin{aligned}
 &g = 9.8 \text{ ms}^{-2} \quad ; \quad V = Er \quad ; \quad P = \vec{F} \cdot \vec{v} \quad ; \quad qE = mg \quad ; \quad A_1 v_1 = A_2 v_2 \quad ; \quad 2gh = v_f^2 - v_i^2 \quad ; \quad mgh = K.E - \text{Work done by air resistance} \\
 &x = v_i t \quad ; \quad F_c = F_g \quad ; \quad eBr = mv_p \quad ; \quad I = nAvQ \quad ; \quad v_f = v_i + gt \quad ; \quad p^2 = \frac{E^2 - m^2 c^4}{c^2} \quad ; \quad k = \frac{1}{4\pi\epsilon_0} = 9 \times 10^9 \text{ Nm}^2 \text{C}^{-2} \\
 &I = \frac{P}{A} \quad ; \quad R = \rho \frac{L}{A} \quad ; \quad E = \frac{kq}{r^2} \quad ; \quad E = \frac{F}{q} \quad ; \quad F = \frac{kq_1 q_2}{r^2} \quad ; \quad y = v_i t + \frac{1}{2} gt^2
 \end{aligned}$$